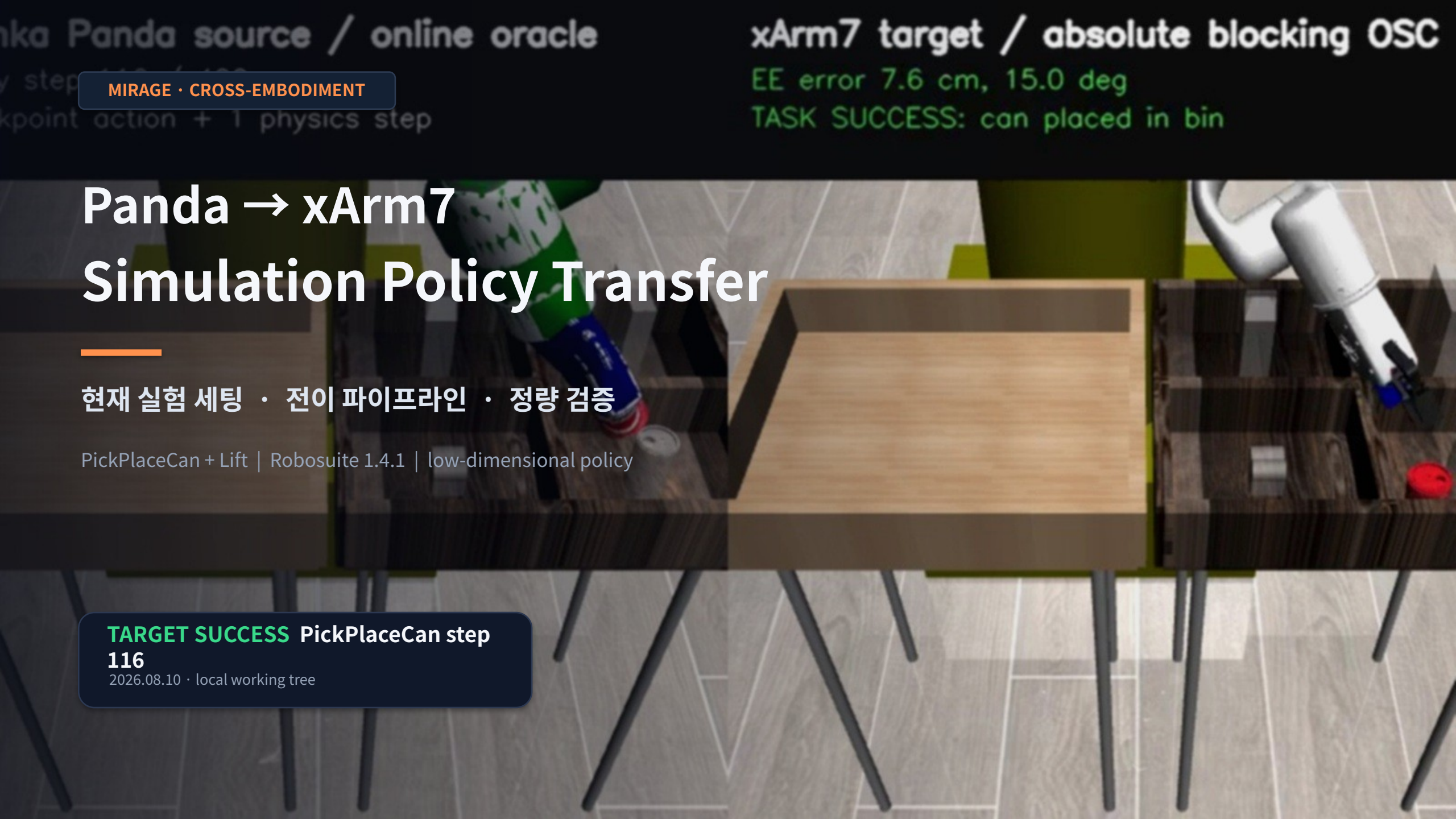


MIRAGE · CROSS-EMBODIMENT

Panda → xArm7 Simulation Policy Transfer

현재 실험 세팅 · 전이 파이프라인 · 정량 검증

PickPlaceCan + Lift | Robosuite 1.4.1 | low-dimensional policy

TARGET SUCCESS PickPlaceCan step 116

2026.08.10 · local working tree

한 줄 결론

정책 실행 주체와 성공 판정 주체를 분리해 xArm7 전이를 검증

정책은 Panda 에서 실행하고 , 성공은 xArm7 에서 판정했다 .

동일한 저차원 체크포인트 출력을 Panda 시뮬레이터에서 1 step 실행한 뒤 ,
달성 pose 를 변환 · 검증하여 xArm7 이 절대 pose 로 추종한다 .

의미 **로봇 형태가 달라도 task-space pose contract 로 실행 경로를 분리**

현재 결과는 실제 로봇이 아닌 시뮬레이션 검증이다 .

1 / 1

PickPlaceCan

target step 116
contact-relaxed steps 32

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Lift

seeds 0, 1, 3, 7
target step 41–57

0

실패 이벤트

timeout 0
iteration limit 0
rejection 0

검증 **현재 작업 트리 회귀 테스트 87 passed**

sources: task-aware rollout reports + pytest

실험 세팅

source policy 와 target control environment 를 완전히 분리

SOURCE

Franka Emika Panda

Robosuite robot id: Panda

- 저차원 robomimic checkpoint
- action: [Δp(3), Δrotvec(3), gripper]
- OSC_POSE · control_delta=true · 20 Hz
- checkpoint action + 1 physics step
- camera / offscreen rendering disabled

panda_sim_world

pose contract

TARGET

xArm7

in-process Robosuite target

- policy camera 를 상속하지 않는 control env
- absolute blocking OSC_POSE
- target task success 로만 early stop
- workspace / joint / freshness safety gate
- gripper: −1 open, +1 closed

xarm7_sim_world

범위 **SIMULATION ONLY** · ROS command / xArm SDK / robot IP / real calibration 사용 안 함

source: xarm_transfer_sim.example.yaml

End-to-end 실행 파이프라인

온라인 oracle 과 learned Ridge 가 같은 계약 · 안전 계층을 공유

01 **Checkpoint**
raw action + metadata

02 **Panda step**
one policy step + achieved pose

03 **Transform v2**
task world × tool frame

04 **Dynamics**
online oracle or Ridge

05 **Safety gate**
fresh · finite workspace · joints

06 **xArm7 OSC**
absolute blocking + outer loop

07 **Target success**
task event terminates

Provider 만 교체 oracle: Panda achieved next pose | Ridge: autoregressive predicted pose

achieved pose feedback

같은 frame·unit·timestamp contract 와 fail-closed validation 을 모든 단계에서 유지

source: rollout.py · contracts.py · runtime.py

좌표 변환 v2

task/world 정합과 tool convention 을 분리해 motion direction 왜곡을 방지

$$T_{targetWorld,targetTool} = T_{targetWorld,sourceWorld} \; @ \; T_{sourceWorld,sourceTool} \; @ \; T_{sourceTool,targetTool}$$

LEFT-MULTIPLY

Task / world transform

Panda 와 xArm7 이 동일한 task 좌표를 사용

현재 값 IDENTITY

```
translation = [0, 0, 0] m
rotation = [0, 0, 0, 1] XYZW
```

RIGHT-MULTIPLY

Tool-frame correction

nominal gripper axis 차이만 local-Z 에서 보정

현재 값 – 90° about local Z

```
quaternion = [0, 0, −0.7071, 0.7071]
```

핵심 home pose 로 변환을 추정하지 않으며 , simulation transform 을 real calibration 으로 표시할 수 없다 .

Target control & safety

absolute blocking OSC 에 보상 루프와 task-aware 예외를 제한적으로 적용

ADMISSION GATES

Frame / type	simulation_only · xarm7_sim_world
Freshness	age ≤ 0.15 s · future skew ≤ 0.02 s
Workspace	[−0.65,−0.60,0.40]–[0.35,0.60,1.35] m
Joint limits	model limit + 0.01 rad margin
Learned model	held-out position / orientation bounds

BLOCKING OSC

Free-space convergence

2.5 cm
position tol.

7° orientation
tolerance

- 5 consecutive in-bound steps
- position correction ≤ 3 cm
- orientation correction ≤ 5°
- timeout 2 s · max 300 iterations

CONTACT EXCEPTION

PickPlaceCan only

simulator 가 gripper ↔ Can 접촉을 직접 확인한 동안만

6.0 cm / 18°

- Kp = [300,300,500,300,300,300]
- position compensation ≤ 10 cm
- Lift 에서는 항상 비활성
- workspace/joint/freshness gate 는 유지

Target task-success event 가 발생하면 마지막 EE pose 가 미수렴이어도 즉시 종료

평가 프로토콜

task profile 이 checkpoint·observation·controller 를 정적으로 고정

TASK 01

PickPlaceCan

- object obs [14] · graspable Can
- checkpoint: can / low_dim / epoch 250
- native horizon 400 · full task seed 1
- **contact relaxation enabled**

TASK 02

Lift

- object obs [10] · graspable cube
- checkpoint: lift / low_dim / epoch 1000
- native horizon 400 · seeds 0,1,3,7
- **contact relaxation disabled**

PROVIDERS

online_oracle
Panda achieved next pose 를 즉시 사용
learned_ridge
recorded action 을 autoregressive replay

DATASET

300 transitions

train: seeds 0,2,4,5 (200)
validation: seeds 1,3 (100)
whole-episode split · seed 11

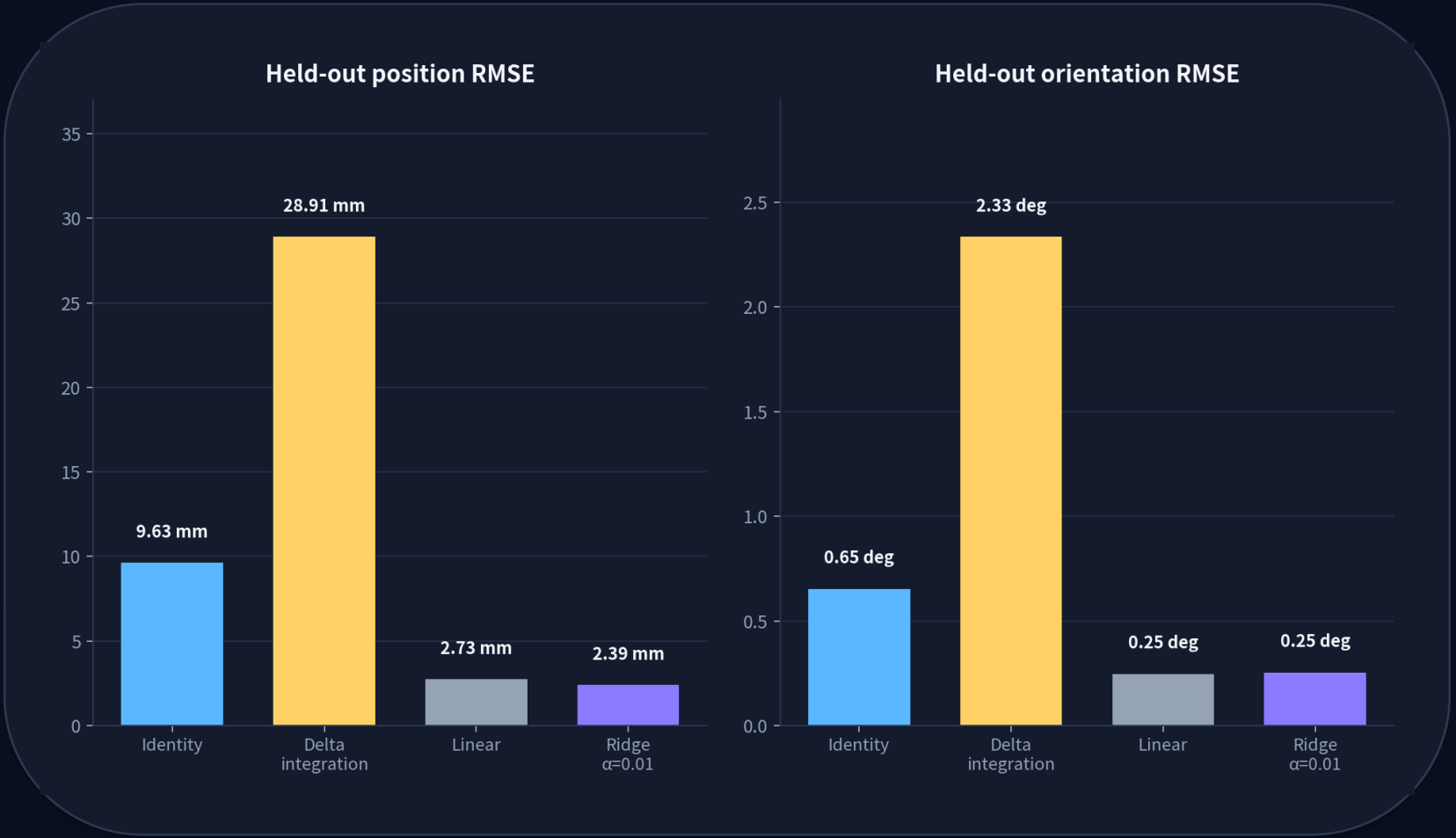
RUNTIME CONTRACT

low_dim 만 실행 가능
image modes: classified, fail closed
unregistered task: fail closed
Lift Ridge training: 현재 범위 밖

source: profiles.py · dataset/model reports

Forward dynamics: held-out one-step prediction

episode leakage 없이 100 개 validation transition 으로 비교



2.39 mm

Ridge position RMSE

α = 0.01 · selected deployable model

0.254°

Ridge orientation RMSE

held-out one-step prediction

≈75% ↓ position RMSE

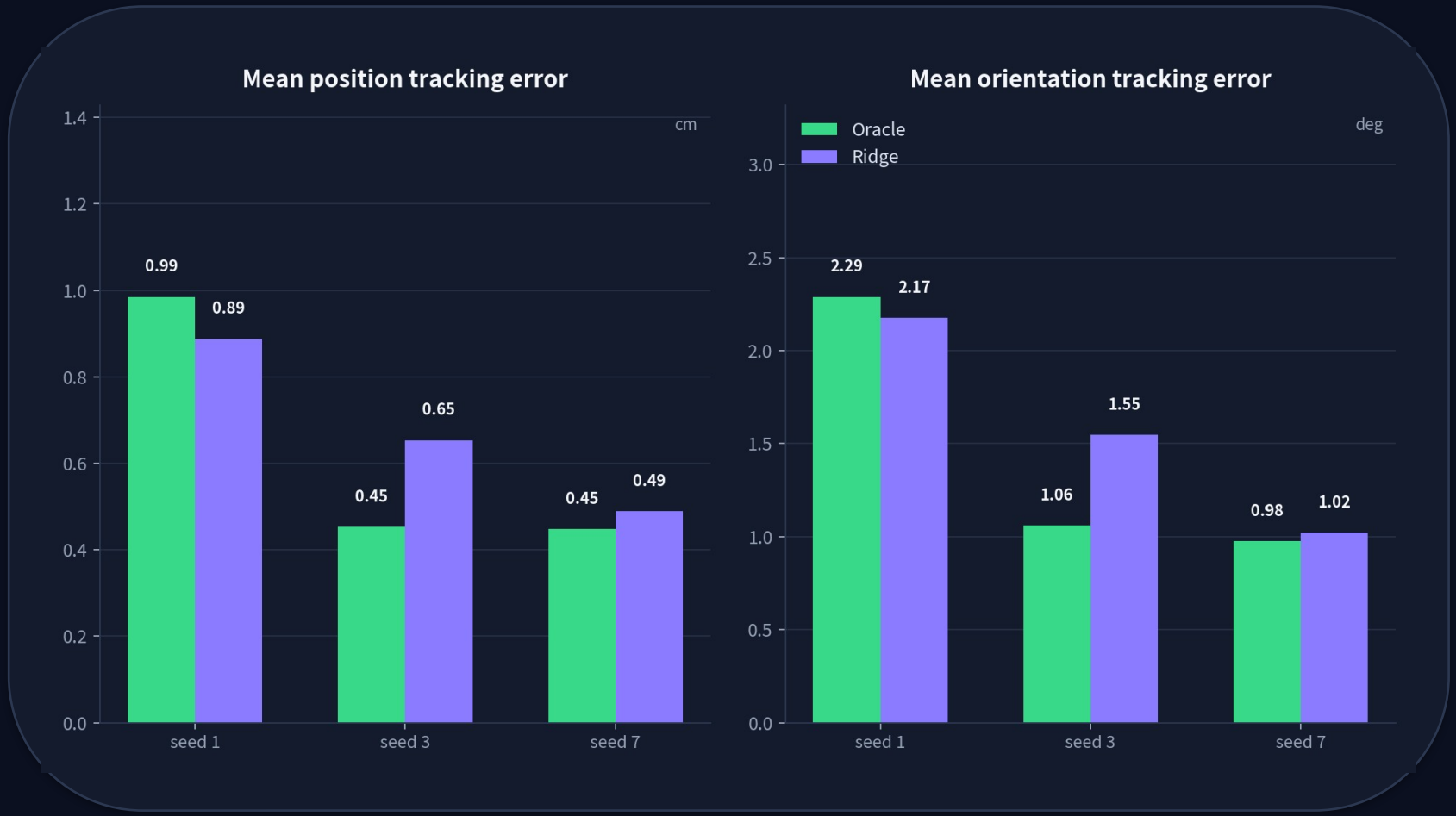
Ridge vs. identity baseline
9.63 mm → 2.39 mm

주의 : one-step held-out error 는 50-step autoregressive drift 의 online safety bound 가 아니다 .

source: forward_dynamics_evaluation.json

50-step xArm7 tracking: Oracle vs. Ridge

동일 held-out seeds 1·3·7, 같은 transform·controller·safety stack



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rollouts complete

Oracle 3 + Ridge 3
all 50 / 50 steps

0

control failures

timeout · iteration limit
rejection · contact event

Ridge 도 target tracking 유지

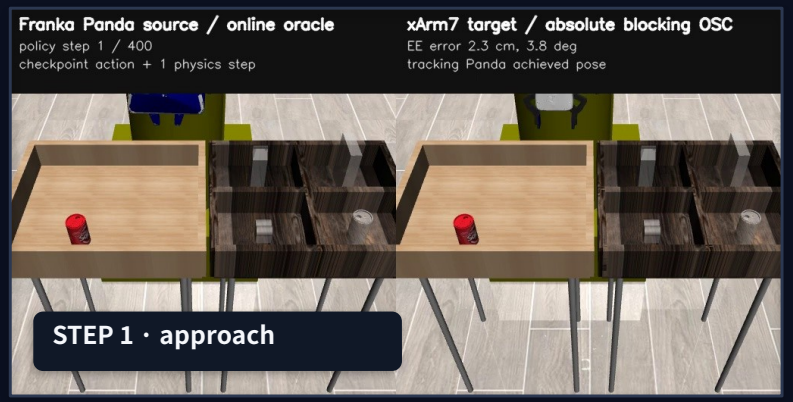
mean pos 0.49–0.89 cm
mean ori 1.02–2.17°

그래프는 controller 가 달성한 target tracking mean error; learned source-pose drift 와 분리해 보고

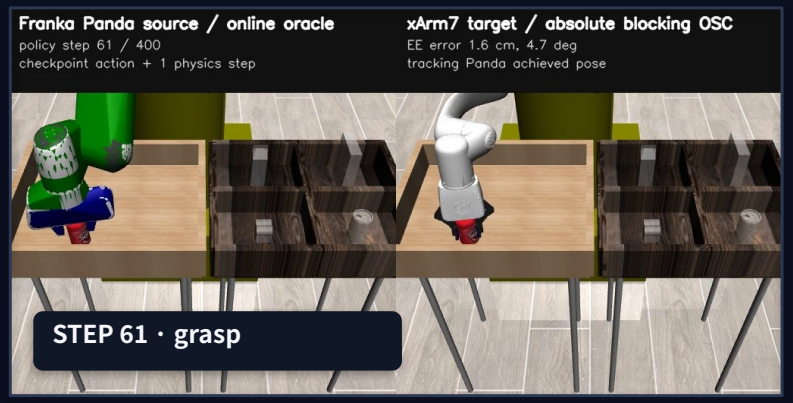
source: task_tool_v2 oracle / learned rollout reports

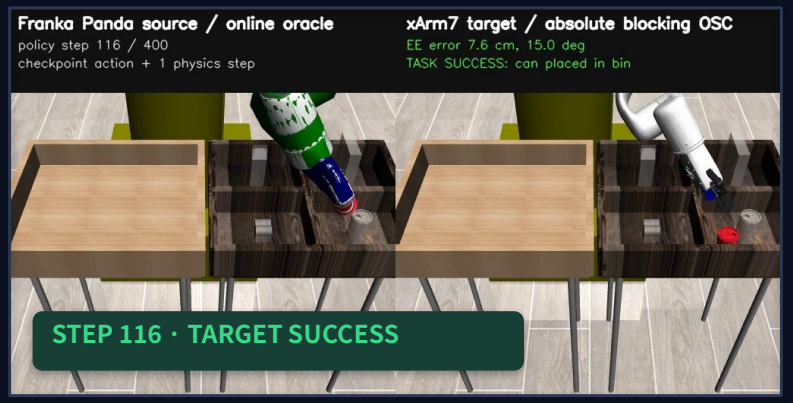
Full-task 결과 ① PickPlaceCan

xArm7 가 can 을 집어 bin 에 놓는 실제 task event 로 성공 판정









SEED 1

116 / 400

policy steps until target success

- timeout 0 · iteration limit 0
- safety rejection 0
- **verified-contact steps 32**
- mean command latency 38.6 ms
- target-only early termination

final task event > pose convergence

성공 시점의 큰 EE residual 은 object contact / immediate task-success 규칙으로 별도 기록됨

source: panda_xarm_oracle_seed1_full_task.mp4 + report

Full-task 결과 ② Lift

모든 seed 에서 target success 까지 계속 실행 ; contact relaxation 은 사용하지 않음



4 / 4

target success

seeds 0 · 1 · 3 · 7

≤ 2.07 cm

maximum position error

all four full-task rollouts

0 failures

timeout · iteration limit
rejection · contact tolerance

seed 1: Panda success step 40 → xArm7 success step 41; source success 가 target rollout 을 조기 종료하지 않음

source: lift/low_dim/xarm_rollout_report.json

재현 방법

task-aware CLI 가 checkpoint 에서 task·mode·native horizon 을 해석

RUN

```
cd mirage
MUJOCO_GL=egl ../.venv/bin/python -m \
  mirage.benchmark.robosuite.xarm_transfer.cli rollout \
  --checkpoint mirage/models/can/low_dim/\
    model_epoch_250_PickPlaceCan_success_1.0.pth \
  --mode oracle --seeds 1 --terminate-on-success
```

VERIFY

```
PYTEST_DISABLE_PLUGIN_AUTOLOAD=1 PYTHONPATH=... \
  python -m pytest -q --import-mode=importlib \
  test_xarm_transfer_*.py test_canonical_policy_checkpoint.py
```

ENVIRONMENT

- Ubuntu 22.04.5 · Python 3.10.12
- Robosuite 1.4.1 · MuJoCo 2.3.2
- ROS 2 Humble installed, unused in rollout
- RTX 5060 Laptop GPU · EGL rendering
- branch: agent/xarm7-development
- HEAD: 4e4429e + local working tree

VALIDATION

87 passed in 70.81 s

current working tree · 2026-08-10
results stay under ignored results/

source: CLI README + local validation

결론 및 다음 단계

현재 성과와 경계 조건을 함께 제시

현재 확보한 것

- **Panda policy → xArm7 control 의 명시적 contract**
- **task/world 와 tool-frame 을 분리한 transform v2**
- **PickPlaceCan 1/1, Lift 4/4 target success**
- **Oracle/Ridge 50-step 6/6 completion**
- **87 개 회귀 · 통합 테스트 통과**

아직 주장하지 않는 것

- 실제 xArm7 hardware 성공
- real-world calibration
- image observation runtime
- Lift learned Ridge model
- Stack/Square/Assembly 지원

NEXT

1. **real calibration 경로를 별도 구현**
2. **low-speed fake → real safety gate**
3. **image observation / cross-painting 연결**
4. **Lift 용 dynamics dataset·Ridge 학습**
5. **multi-seed PickPlaceCan 확대**

Takeaway · embodiment 차이는 pose contract 로 흡수하고 , 안전 · 성공 판정은 target 에서 닫는다 .